PINNGraPE

Physics-Informed Neural Network for Gravitatioanal-wave Parameter Estimation

Matteo Scialpi, Michał Bejger, Francesco Di Clemente

Physics and Earth Science Department - Ferrara University

Istituto Nazionale di Fisica Nucleare (INFN) – Ferrara

This project was partially financed by PRIN 202275HT58

CoCoNut meeting, Valencia, ES

December 13th, 2024

1. Introduction
2. Architecture
3. Results
4. Conclusions

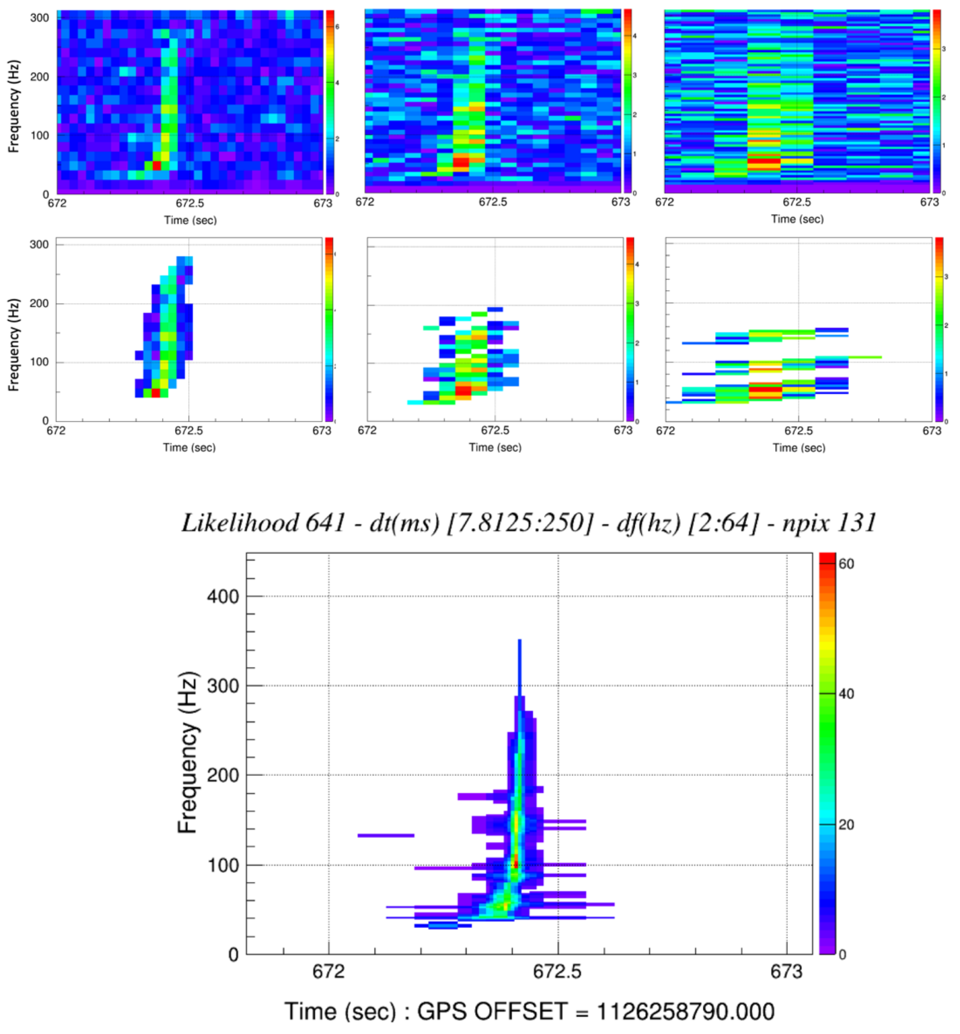
Introduction

- **PINNGraPE** is a ML (Machine Learning) algorithm which **performs PE** (Parameter Estimation) **for a GW signal's source thanks to a PINN** (Physics-Informed Neural Network) [1].

- **Why?**
  - **cWB [2] needs a PE component for CBCs** (Compact Binary Coalescences). It is an unmodelled search: **needs for a theoretical physics information.**

Introduction: cWB

- **cWB is an unmodelled** pipeline used in modern interfeometers by LVK collaboration.
- **Unmodelled:** based on pure coherence of energy pixels in the frequency domain
- No theoretical background for sources: **no PE on the source is possible!**



[2]

Introduction: Physics-Informed Neural Networks

- PINNs are NNs originally developed in fluid dynamics to **solve or discover differential equations behind physical data.**
- **Base of the physics information:** physics-informed loss function

$$u_t + \mathcal{N}[u; \lambda] = 0,\ x \in \Omega,\ t \in [0, T],$$

$$f := u_t + \mathcal{N}[u],$$

$$MSE = MSE_u + MSE_f,$$

where

$$MSE_u = \frac{1}{N_u} \sum_{i=1}^{N_u} |u(t_u^i, x_u^i) - u^i|^2,$$

and

$$MSE_f = \frac{1}{N_f} \sum_{i=1}^{N_f} |f(t_f^i, x_f^i)|^2.$$

Task: find an approximation for u (= solve the differential equation) and optimize λ.

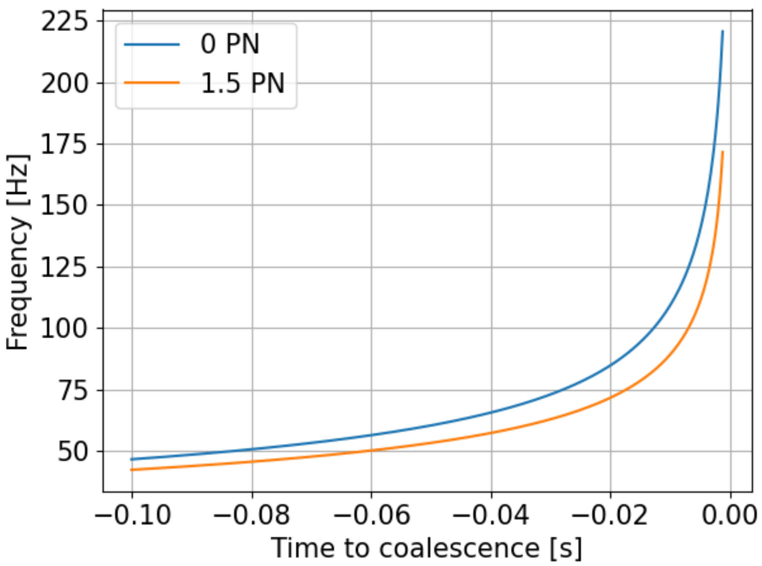
Data: (u^i, t^i,x^i).

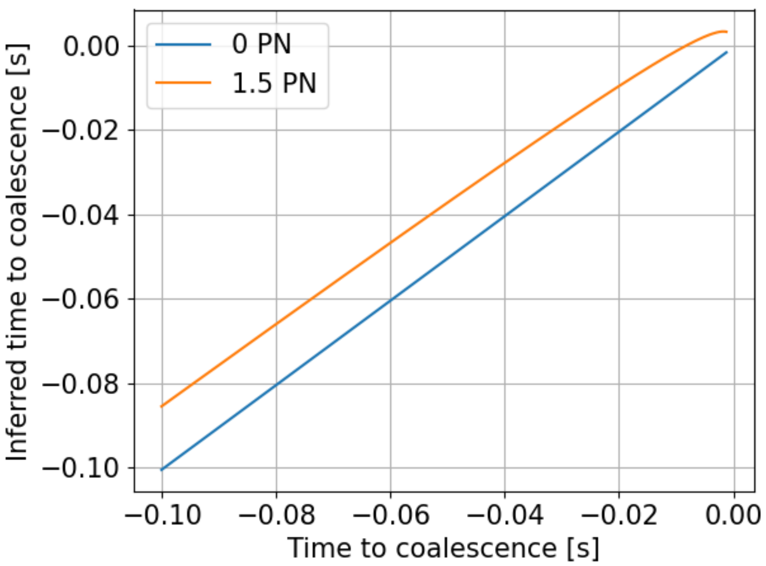
How: approximate u with a NN and minimize the loss.

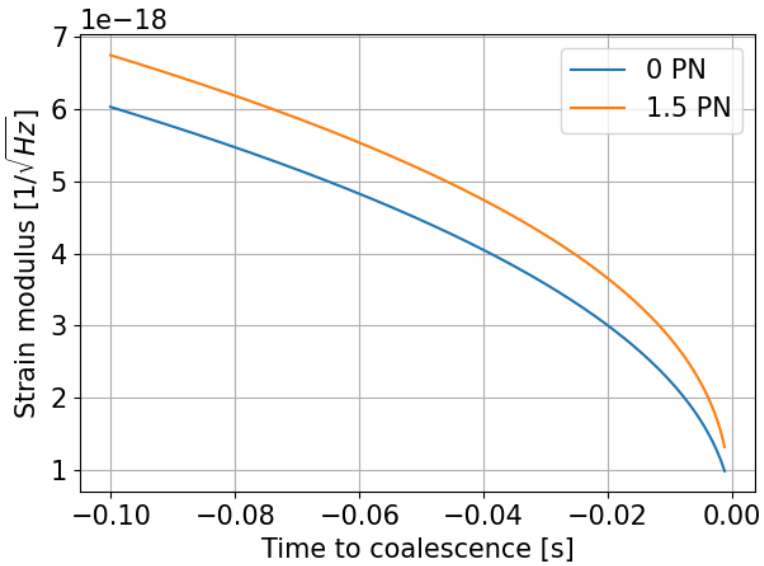
Why this should work? Since u is a NN, f is a PINN: they share the same parameters.

Architecture: training dataset

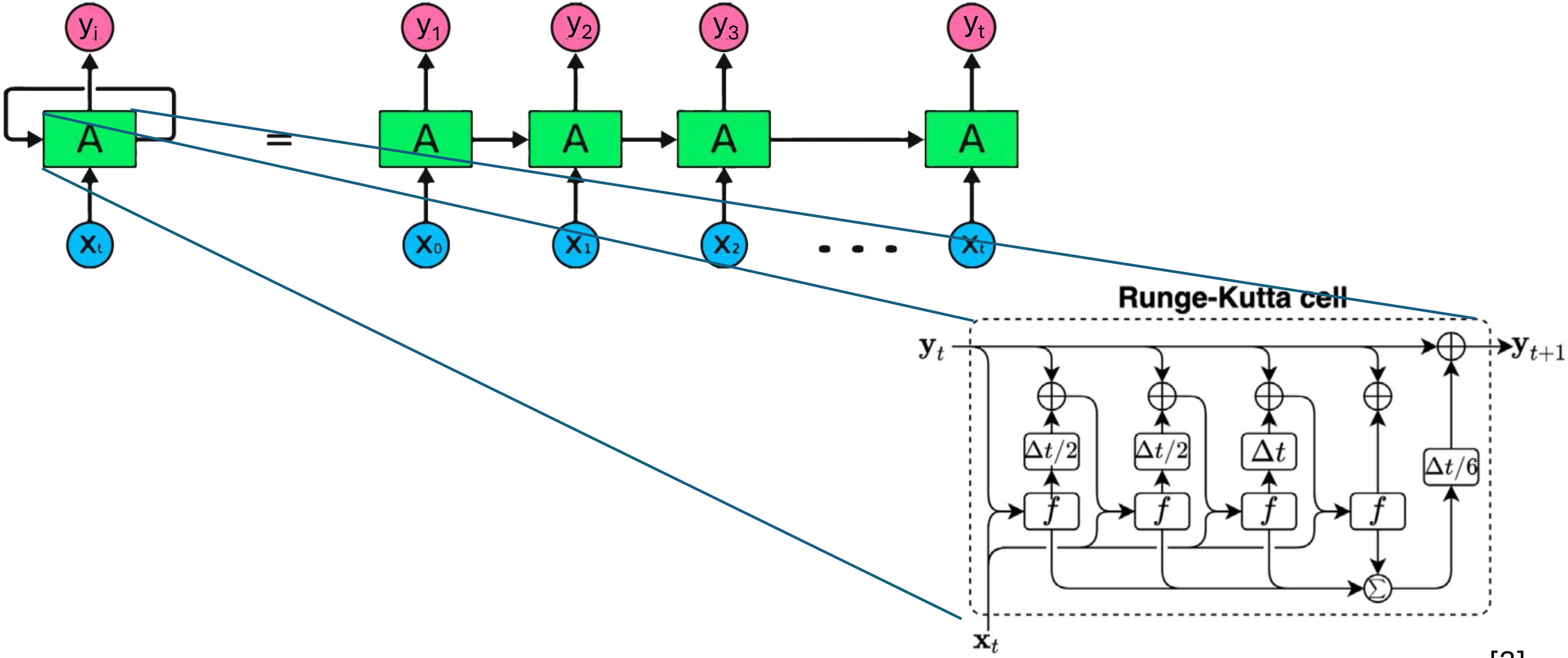
Data: $\{(t_k, f_k), (t_k, t'_k), (t_k, |\tilde{h}_k|)\}$ points,
where $t'_k = t[f(t_k)]$ and $\tilde{h}_k = \tilde{h}[f(t_k)]$.







Architecture: Recurrent Neural Network



[3]

Architecture: Workflow

Task: infer η and Mtot behind the data.

1. Impose the following:

$$\frac{df}{dt} - \mathcal{F}[f, \eta, M_{tot}] = g; \qquad g = 0 \qquad (1)$$

where F is a known functional.

2. Solve (1) thanks to the RNN to get $f(t_k)$.

3. Calculate $t(f(t_k))$ and $h(f(t_k))$.

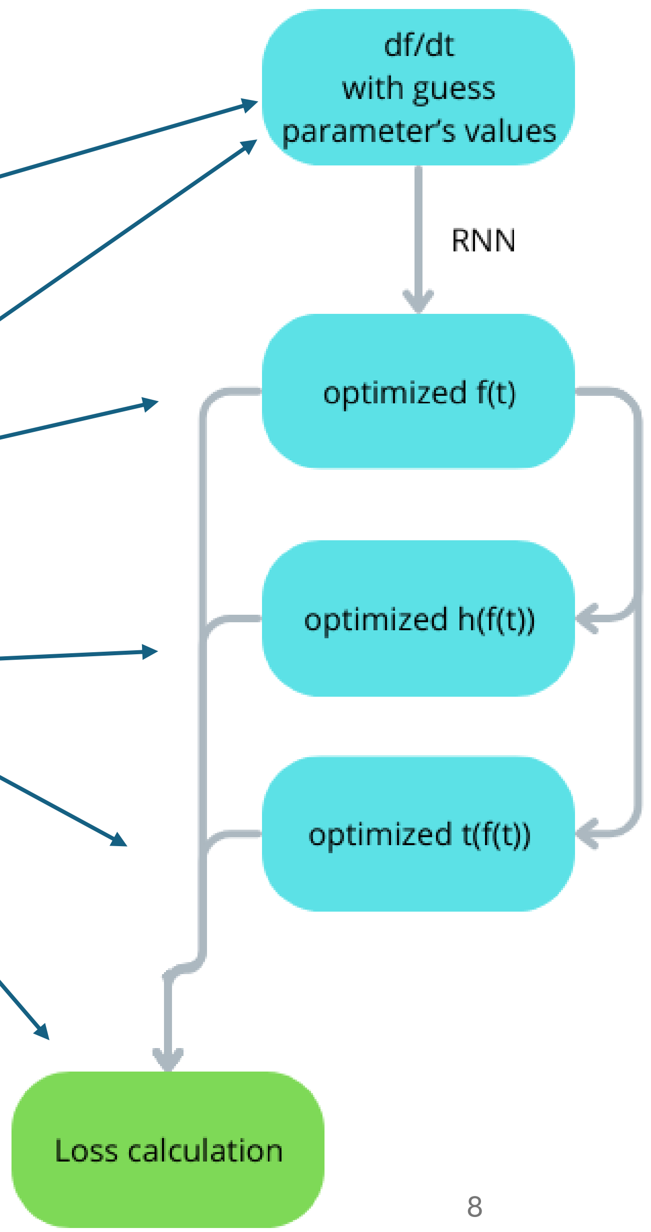
4. Calculate the loss:

$$\mathcal{L} = \frac{\beta_f}{N}\sum_{k=1}^{N}|f_k - f(t_k)| + \frac{\beta_t}{N}\sum_{k=1}^{N}|t_k - t(f(t_k))| + \frac{\beta_h}{N}\sum_{k=1}^{N}|h_k - h(f(t_k))|$$

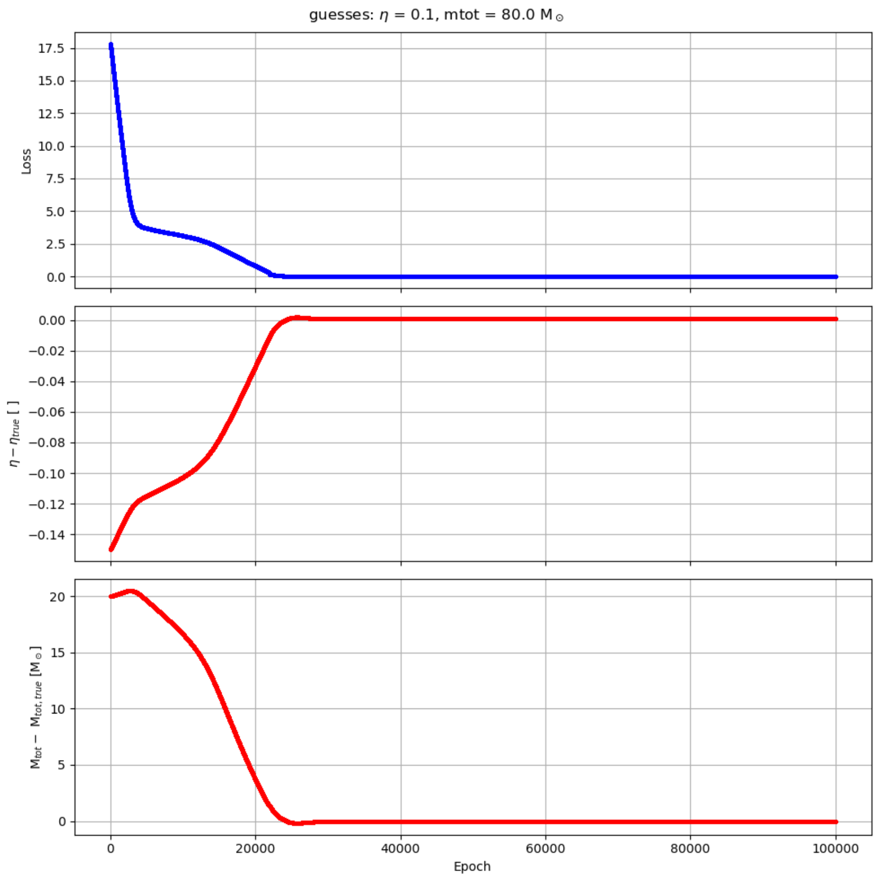
5. Optimize parameters.

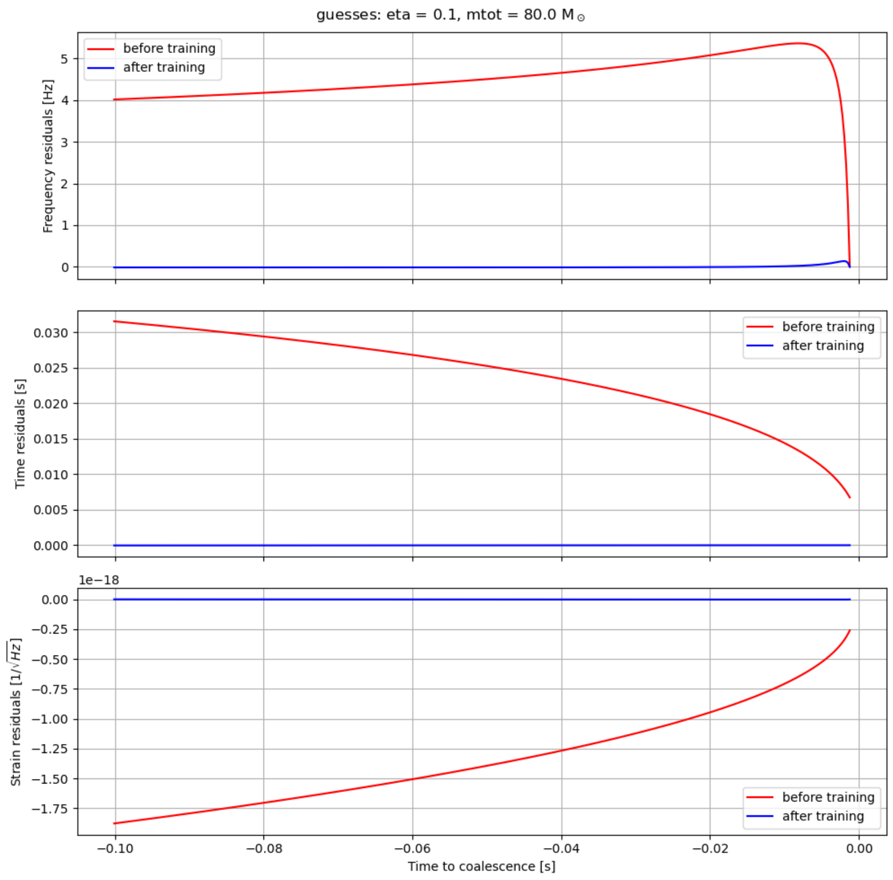
6. Repeat 2-5 until convergence.

Learning algorithm



Results - BBH

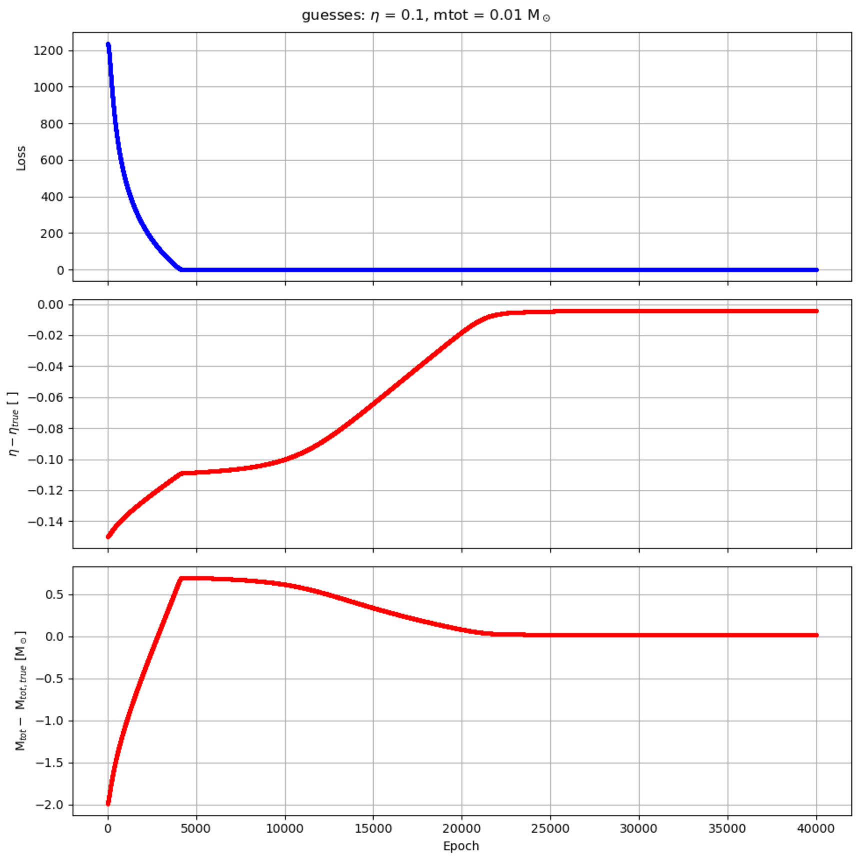


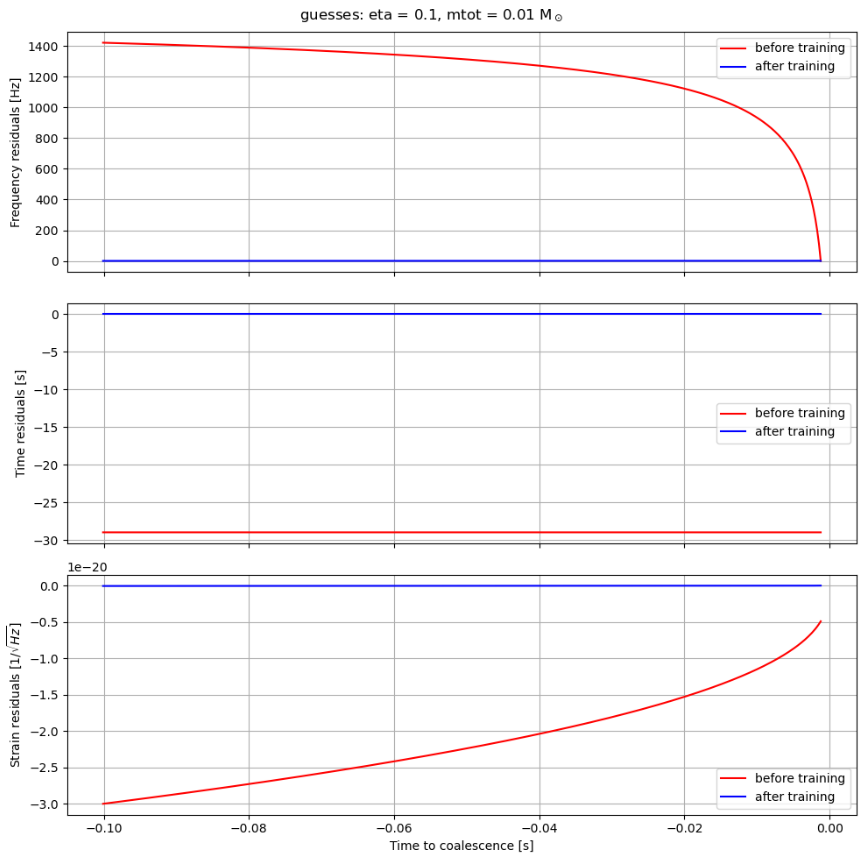


- Final loss value: 2.385 * 10^(-2).
- AdamW optimizer
- Total time: -1h on CPU (4000 epochs).

	η	M_{tot} $[M_\odot]$
Guess	**0.1000**	**80.00**
Target	**0.2500**	**60.00**
Inferred	**0.2507**	**59.95**

Results - BNS





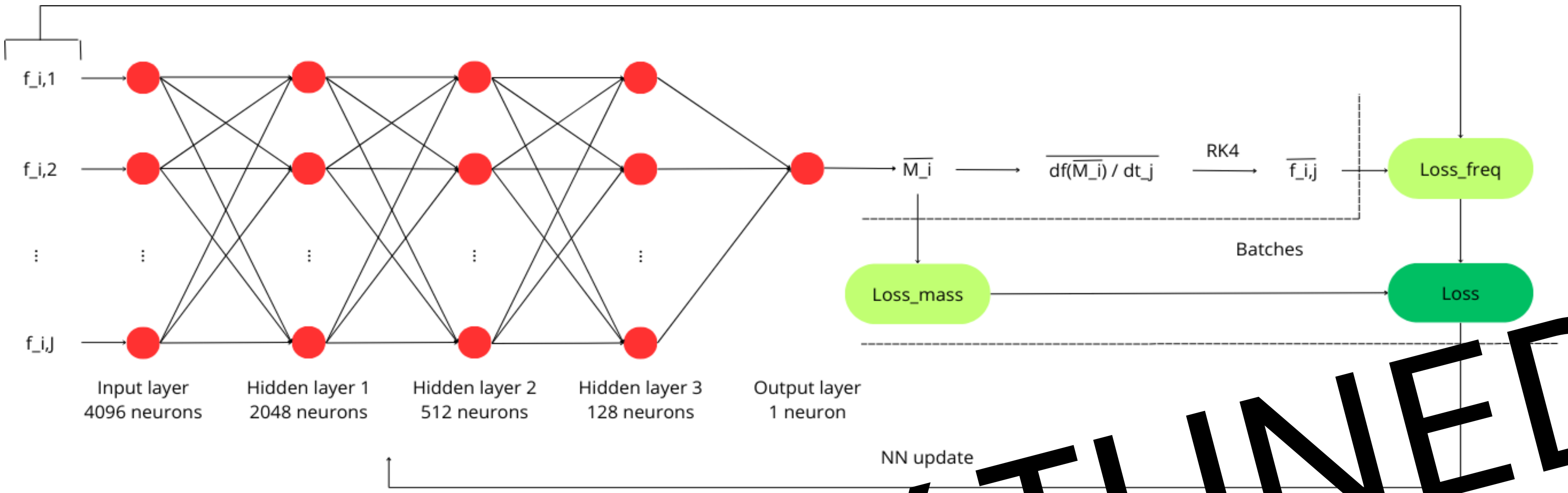
- Final loss value: 5.899 * 10^(-2).
- AdamW optimizer
- Total time: -1h on CPU (4000 epochs).

	η	M_{tot} [$M_\odot$]
Guess	0.1000	0.0100
Target	0.2500	2.000
Inferred	0.2459	2.018

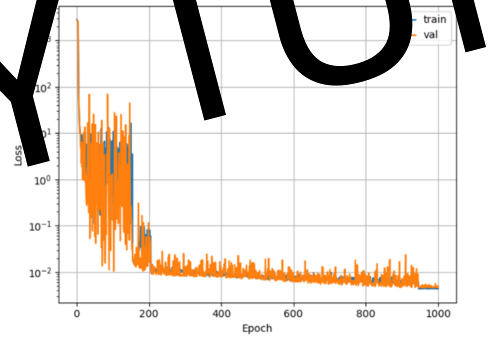
Conclusions

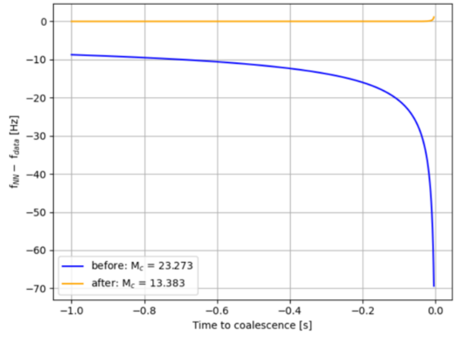
- **PINNGraPE is able to infer η and Mtot values with 10^{-2} relative error** from f, t and h data, introducing 1.5PN formalism.

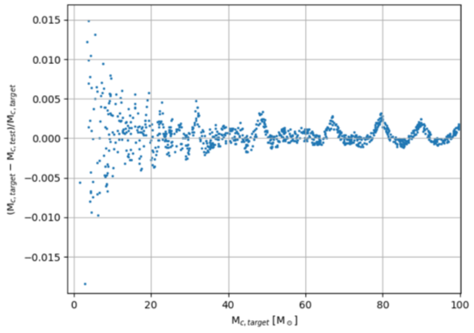
- Next steps:
  - optimize the architecture to handle a training dataset with a parameter range for η and Mtot (LSTM, minLSTM, Autoencoder and more);
  - use cWB real outputs (WIP).

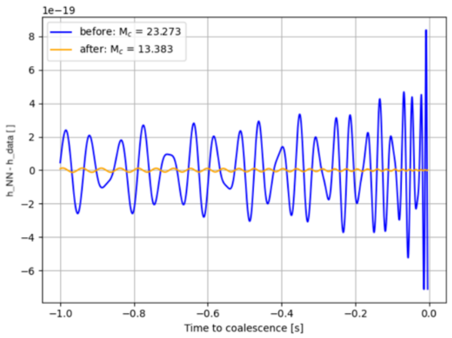

f_i,1
f_i,2
f_i,J
Input layer
4096 neurons
Hidden layer 1
2048 neurons
Hidden layer 2
512 neurons
Hidden layer 3
128 neurons
Output layer
1 neuron
M_i
df(M_i) / dt_j
RK4
f_i,j
Loss_freq
Batches
Loss_mass
Loss
NN update

STAY TUNED


train
val
Loss
Epoch


before: Mc = 23.273
after: Mc = 13.383
Time to coalescence [s]




1e−19
before: Mc = 23.273
after: Mc = 13.383
Time to coalescence [s]

References

[1] M. Raissi, P. Perdikaris, and G. Karniadakis, “Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations,” Journal of Computational Physics, vol. 378, pp. 686–707, 2019.

[2] M. Drago, S. Klimenko, C. Lazzaro, E. Milotti, G. Mitselmakher, V. Necula, B. O’Brian, G. A. Prodi, F. Salemi, M. Szczepanczyk, S. Tiwari, V. Tiwari, G. V, G. Vedovato, and I. Yakushin, “coherent waveburst, a pipeline for unmodeled gravitational-wave data analysis,” SoftwareX, vol. 14, p. 100678, June 2021.

[3] R. G. Nascimento, K. Fricke, and F. A. Viana, “A tutorial on solving ordinary differential equations using python and hybrid physics-informed neural network,” Engineering Applications of Artificial Intelligence, vol. 96, p. 103996, 2020

[4] C. Cutler and E. E. Flanagan, “Gravitational waves from merging compact binaries: How accurately can one extract the binary’s parameters from the inspiral waveform?,” Physical Review D, vol. 49, p. 2658–2697, Mar. 1994.

[5] the LIGO Scientific Collaboration and the Virgo Collaboration, “The basic physics of the binary black hole merger gw150914,” Annalen der Physik, vol. 529, Oct. 2016.